Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | High-Precision, Intrinsically Polarization-Selective Photon Detectors Enabled by Scalable Chiral Nanomaterials | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Electrical · Electronics | | Mechanical · Materials | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Optical Application Devices | Optical Materials | Nano/Micro Mechanical Systems | Nano/Micro Sensors |
| Concept of Proposed Technology | This project aims to develop a next-generation polarization-selective photon detection platform by embedding polarization selectivity directly into scalable chiral nanomaterials, rather than relying on external polarization optics. Conventional polarization detection systems typically depend on bulky optical components such as linear or circular polarizers, waveplates, and analyzers, combined with multi-step intensity measurements. These approaches impose fundamental limitations on sensor miniaturization, alignment tolerance, mechanical robustness, and array scalability.<br><br>In contrast, the proposed technology leverages the intrinsic asymmetric light–matter interactions of chiral metallic or semiconductor nanostructures, enabling polarization selectivity to be encoded at the material and structural level. As a result, polarization states can be transduced directly into electrical signals at the single-pixel level, allowing single-shot polarization discrimination without external polarizers or complex optical alignment. This materials-encoded approach not only supports compact and pixelated detector architectures but also enhances measurement precision by reducing susceptibility to optical drift, misalignment, and system-level noise.<br><br>The resulting polarization-selective photonic detector platform is broadly applicable to polarization-resolved imaging, machine vision, optical communication, and photonic security systems, where polarization carries critical information beyond intensity alone. Additional application domains include biomedical and chemical sensing, where chiral or anisotropic light–matter interactions provide | | | |

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| | enhanced contrast, as well as emerging photonic and quantum-optical systems that require reliable and scalable detection of polarization states. |
| **Rationale for Joint R&D Project** | To advance chiral nanophotonic concepts to TRL 4–5 device-level polarization-selective detector modules, several technical elements must be integrated simultaneously, posing a significant challenge for a single institution.<br><br>First, scalable fabrication and reproducibility of chiral nanomaterials and nanostructures are required, including wafer-scale or large-area patterning, batch-to-batch uniformity, and compatibility with pixelated detector arrays. Second, these chiral nanostructures must be integrated into practical photodetector architectures (e.g., photodiodes, photoconductors, hybrid or CMOS-compatible platforms), requiring careful design of electrodes, interfaces, passivation layers, packaging, and electrical readout schemes. Third, polarization-selective performance must be validated using quantitative and system-relevant metrics, including extinction ratios, degree-of-polarization accuracy, crosstalk, drift, and long-term operational stability.<br><br>The proposing research group (Kim group) at RPI has strong expertise in the design, patterning, and optical engineering of chiral metallic and semiconductor nanostructures [1-10], with a particular focus on CPL-selective optical responses and their physical mechanisms [11,12]. However, translating these materials-level innovations into device-level CPL or polarization-selective photodetectors, including scalable fabrication, array integration, packaging, and reliability testing, poses practical limitations in terms of infrastructure and industrial process experience.<br><br>Collaboration with a Korean SME possessing strengths in photodetector fabrication, device integration, and manufacturing-oriented validation provides a highly complementary partnership. Through this joint R&D framework, RPI will focus on chiral nanomaterial and polarization-selective optical design, while the Korean SME will lead device-level integration, module development, and system-oriented validation. This division of expertise enables efficient resolution of TRL bottlenecks and accelerates the transition from materials innovation to deployable polarization-selective detector technologies. |
| **Key Words** | 1. Chiral optical nanomaterial<br>2. Photonic sensing<br>3. Material-encoded polarization selectivity<br>4. Polarization-resolved imaging |

## 3. Research and Development Objectives & Plan

| **Technology Readiness Level** (TRL)[2] | **Start of the Project** | Level (2-3) | **End of the Project** | Level (4-5) |
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| **R&D Objectives** | Polarization provides an additional information dimension beyond intensity, enabling sensitivity to anisotropy, surface morphology, stress, crystallinity, and chiral light–matter interactions. Despite its importance, practical polarization detection in compact systems remains constrained by reliance on external polarization optics, multi-step measurements, and alignment-sensitive architectures, which limit precision, scalability, and long-term stability.<br><br>The central objective of this project is to develop polarization-selective photodetectors in which polarization discrimination is intrinsically encoded in chiral nanomaterials, enabling direct electrical transduction of polarization states at the single-pixel level. By shifting polarization selectivity from system-level optical components to material-level responses, the proposed approach aims to achieve high-precision polarization detection that is robust against optical misalignment, drift, and environmental variation.<br><br>Rather than defining precision solely in terms of signal-to-noise ratio, this project emphasizes | | | |

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| | polarization fidelity, reproducibility, and stability as key performance metrics. Through device-level integration and quantitative validation, the project seeks to establish a scalable and manufacturable foundation for polarization-selective photodetectors suitable for real-world photonic sensing and imaging applications. |
| **R&D Target Specifications** | 1) Polarization Selectivity and Chiroptical-to-Electrical Performance<br>• CPL selectivity metric (e.g., $g_{ph} = (I_L - I_R)/(I_L + I_R)$): ≥ 0.10<br>• CPL extinction ratio (LCP vs. RCP): ≥ 10:1<br>• Linear polarization option: DoLP estimation error ≤ 0.05<br>• Inter-pixel or inter-channel polarization crosstalk: ≤ 10%<br>2) Photodetector Core Performance<br>• Responsivity: ≥ 0.10 A/W<br>• Specific detectivity (D*): ≥ 1 × $10^{11}$ Jones<br>• Dark current density: ≤ 1 × $10^{-6}$ A/cm²<br>• Bandwidth: ≥ 10 kHz (imaging/sensing target)<br>  ○ Optional communication-oriented track: ≥ 1 MHz<br>3) Stability and Reliability<br>• Retention of polarization selectivity: ≥ 90% after ≥ 4 weeks<br>• Thermal stability: performance variation ≤ 10% from 0–60 °C<br>• Photostability: degradation ≤ 10% after defined optical exposure<br>4) Array and System Demonstration<br>• Demonstration of a minimum 4 × 4 pixel array (or equivalent)<br>• Polarization state classification accuracy: ≥ 90%<br>• Demonstration of calibration and correction workflows for DoCP/DoLP estimation |
| **R&D Plan** | **Aim 1. Joint System Definition and Performance Metrics (0–6 Months)**<br>• Selection of priority application scenarios<br>• Definition of target wavelength range and detector architecture<br>• Standardization of polarization performance metrics and validation protocols<br>• IP strategy and data-exchange framework<br>Deliverables: System requirements document, validation protocol v1, IP roadmap<br><br>**Aim 2. Scalable Chiral Nanomaterials with Encoded Polarization Selectivity (0–12 Months)**<br>• Design and fabrication of scalable chiral nanostructures<br>• Identification of structure–property relationships governing CPL selectivity<br>• Evaluation of batch-to-batch reproducibility<br>Deliverables: 2–3 candidate structures, optical datasets, reproducibility report<br><br>**Aim 3. Device Integration and Single-Pixel Demonstration (7–24 Months)**<br>• Integration of chiral nanostructures with photodetector architectures<br>• Optimization of electrical performance and environmental stability<br>• Demonstration of CPL-selective photodetection at the device level<br>Deliverables: Single-pixel device prototypes, performance benchmark matrix<br><br>**Aim 4. Array-Level Integration and Polarization Calibration (19–30 Months)**<br>• Fabrication of small-scale detector arrays<br>• Crosstalk analysis and polarization calibration workflows<br>• Application-oriented demonstration measurements<br>Deliverables: Array prototypes, calibration protocol v2, demonstration datasets<br><br>**Aim 5. Reliability Testing and Pre-Commercial Prototyping (25–36 Months)**<br>• Long-term stability and environmental reliability testing<br>• Packaging and module-level integration<br>• Final device datasheets and commercialization roadmap<br>Deliverables: Prototype module, datasheet, TRL 4–5 validation package, patent draft |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The Research Institute (RPI) will focus on the upstream materials and optical engineering aspects of the project, where fundamental design choices determine polarization selectivity and device-level performance. In particular, RPI's role is centered on establishing scalable chiral nanostructure designs and quantitative optical benchmarks that can be reliably transferred into photodetector fabrication processes.<br>• Design and scalable patterning of chiral nanostructures<br>• Engineering and characterization of CPL-selective optical responses<br>• Definition of polarization performance metrics and analysis methodologies<br>• Structural and process-tolerance guidelines for device integration |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will lead the downstream device development and system integration efforts, translating chiral nanomaterial designs into practical polarization-selective photodetectors and functional modules. This role emphasizes manufacturability, reliability, and application relevance, ensuring that the developed technology can be validated at the device and system levels.<br>• Photodetector device architecture and process integration<br>• Array fabrication, electrical readout, and calibration workflows<br>• Reliability testing and design-for-manufacturability<br>• Application-driven system demonstrations and commercialization strategy |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | From a technology development perspective, this project establishes a design framework and validation methodology that directly connects the optical responses of chiral nanostructures to electrical polarization readout through a *material-encoded polarization selectivity* approach. By relaxing the conventional dependence of polarization measurements on external optical components, the proposed work enables polarization states, particularly CPL and related polarization information, to be accessed as a stable and independent analytical dimension, distinct from intensity. This capability provides new device-level physical insight and contributes to the development of standardized performance metrics at the intersection of nanophotonics, chiral optics, and sensor engineering.<br><br>From an industrial perspective, the proposed technology reduces or eliminates bulky polarization components such as polarizers and waveplates that introduce size, alignment, and cost constraints. As a result, it enables simultaneous improvements in the following areas:<br>• Compact module integration, supporting miniaturized sensor and detector platforms<br>• Mechanical robustness, with reduced sensitivity to shock, vibration, and alignment drift<br>• Scalability toward array- and camera-level implementations, enabling polarization-resolved imaging systems<br>• Compatibility with manufacturing-oriented fabrication processes, facilitating translation to volume production<br>These advantages create clear entry points into high-value application domains, including polarization-resolved imaging and machine vision (e.g., defect inspection and material identification), polarization-multiplexed optical communications, photonic security systems based on polarization-encoded authentication, and bio-chemical sensing where chiral contrast provides enhanced sensitivity. In addition, this project provides a concrete pathway for the Korean SME partner to develop product-ready polarization detection modules—such as sensors, cameras, or communication receivers—targeting global markets. |

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| | Expected Quantitative Outcomes:<br>• Q1 journal publications: ≥ 2–3<br>• International conference presentations or demonstrations: ≥ 2<br>• Patent applications: ≥ 1 (KR and/or US)<br>• Prototype datasheet and performance validation package: 1 set, supporting TRL 4–5 |

## 6. References


(1) W.J. Choi, J.-Y. Kim M. Hwang, C.M. Lim, B.C Park, A Emre, S. H. Lee, N. A. Kotov, "Highly Elliptic Circular Dichroism of Copper Aspartate One-Dimensional Nanostructures across the Ultraviolet to Terahertz Ranges" ***Nano Letters*****,** 2025, 25, 7699-7706

(2) J. Lu, H. J. Jung, J.-Y. Kim, N. A. Kotov, "Bright, circularly polarized black-body radiation from twisted nanocarbon filaments" ***Science*****,** 2024, 386, pp1400-1404

(3) J.-Y. Kim, C. McGlothin, M. Cha, Z. J.Pfaffenberger, E. S. Turali-Emre, S. Lee, W. Choi, J. Biteen, N. A. Kotov, "Direct-write 3D printing of plasmonic nanohelicoids by circularly polarized light", ***Proceedings of the National Academy of Sciences*** 2024**,** 121, e2312082121

(4) P. Kumar, T. Vo, M. Cha, A. Visheratina, J.-Y. Kim, W. Xu, J. Schwartz, A. Simon, D. Katz, E. Marino, W. J. Choi, S. Chen, C. Murray, R. Hovden, S. Glotzer, N. A. Kotov, "Photonically Active Bowtie Nanoassemblies with Chirality Continuum" ***Nature*** 2023, 615, 418-424

(5) S. Zhou, J. Li, J. Lu, H. Liu, J.-Y. Kim, A. Kim, L. Yao, Ch. Liu, C. Qian, Z. D. Hood, X. Lin, W. Chen, T. E. Gage, I. Arslan, A. Travesset, K. Sun, N. A. Kotov, Qian Chen, "Chiral Assemblies of Pinwheel Superlattices on Substrates", ***Nature*** 2022, 612, 259–265

(6) J. Guo, J.-Y. Kim, S. Yang, J. Xu, Y. C. Choi, A. Stein, C. B. Murray, N. A. Kotov, Cherie R Kagan, "Broadband Circular Polarizers via Coupling in 3D Plasmonic Meta-Atom Arrays", ***ACS Photonics*** 2021, 8, 1286-1292

(7) J. Ahn, S. Ma, J.-Y. Kim, W. Yang, N. A. Kotov, and J. Moon. "Chiral 2D Organic Inorganic Hybrid Perovskite with Circular Dichroism Tunable Over Wide Wavelength Range", ***J. Am. Chem. Soc.*** 2020, 142, 4206–4212

(8) J. Yan, W. Feng, J.-Y. Kim, Z. Mu, P. Kumar, X. Mao, X. Wu, N. A. Kotov, "Self-Assembly of Chiral Nanoparticles into Semiconductor Helices with Tunable Near-Infrared Optical Activity", ***Chemistry of Materials,*** 2020, 32, 476–488

(9) J. -Y. Kim, J. Yeom, H. Calcaterra, G. Zhao, J. Munn, P. Zhang, N. A. Kotov, "Assembly of Gold Nanoparticles into Chiral Superstructures Driven by Circularly Polarized Light", ***J. Am. Chem. Soc.*** 2019, 141, 11739−11744

(10) W. Feng, J.-Y. Kim, X. Wang, H. A. Calcaterra, Z. Qu, L. Meshi, N. A. Kotov, "Assembly of mesoscale helices with near-unity enantiomeric excess and light-matter interactions for chiral semiconductors", ***Science Advances*** 2017, 3, e1601159**.**

(11) L. Ohnoutek, J.-Y. Kim, J. Lu, B. J. Olohan, D. M. Răsădean, G. D. Pantoș, N. A. Kotov, V. K. Valev, "Third Harmonic Mie Scattering from Semiconductor Nanohelices", ***Nature Photonics*** 2022, 16, 126-133

(12) W. Choi, K. Yano, M. Cha, F. Colombari, J. -Y. Kim, Y. Wang, S. H. Lee, J. Kruger, André Farias de Moura, Nicholas A Kotov, "Chiral Phonons in Microcrystals and Nanofibrils of Biomolecules" ***Nature Photonics*** *2022*, 16, 366-373

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |